

10 Oops – egg drop!

Next add the `egg_dropped()` function after `move_eggs()`. When an egg is dropped, it is removed from the list of eggs and then deleted from the canvas. A life is deducted using the `lose_a_life()` function, which you'll create in Step 11. If losing a life means there are no lives left, the "Game Over!" message is shown.

If no lives are left, tell the player that the game is over.

```
root.after(egg_speed, move_eggs)

def egg_dropped(egg):
    eggs.remove(egg)
    c.delete(egg)
    lose_a_life()
    if lives_remaining == 0:
        messagebox.showinfo('Game Over!', 'Final Score: ' \
                               + str(score))
        root.destroy()
```

The egg is removed from the eggs list.

The egg disappears from the canvas.

This line calls the `lose_a_life()` function.

The game ends.

11 Lose a life

Losing a life simply involves subtracting a life from the `lives_remaining` variable and then displaying the new value on the screen. Add these lines after the `eggs_dropped()` function.

```
root.destroy()

def lose_a_life():
    global lives_remaining
    lives_remaining -= 1
    c.itemconfigure(lives_text, text='Lives: ' \
                    + str(lives_remaining))
```

This variable needs to be global, as the function will modify it.

The player loses a life.

12 Check for a catch

Now add the `check_catch()` function. An egg is caught if it's inside the arc of the catcher. To find out if you've made a catch, the `for` loop gets the coordinates of each egg and compares them with the catcher's coordinates. If there's a match, the egg is caught. Then it's deleted from the list, removed from the screen, and the score is increased.

This line updates the text that shows the remaining lives.

```
c.itemconfigure(lives_text, text='Lives: ' + str(lives_remaining))

def check_catch():
    (catcher_x, catcher_y, catcher_x2, catcher_y2) = c.coords(catcher)
    for egg in eggs:
        (egg_x, egg_y, egg_x2, egg_y2) = c.coords(egg)
        if catcher_x < egg_x and egg_x2 < catcher_x2 and catcher_y2 - egg_y2 < 40:
            eggs.remove(egg)
            c.delete(egg)
            increase_score(egg_score)
    root.after(100, check_catch)
```

Get the coordinates of the catcher.

Get the coordinates of the eggs.

Is the egg inside the catcher horizontally and vertically?

Increase the score by 10 points.

Call this function again after 100 milliseconds (one-tenth of a second).